

Countable vs Uncountable Sets: Complete Reference

COUNTABLE SETS (Can be listed in a sequence)

Finite Sets (All are countable)

1. Integers less than 100
2. Positive integers less than 1,000,000,000
3. Integers with absolute value less than 1,000,000
4. Any finite subset of any set

Countably Infinite Sets

1. **Natural numbers** ($\mathbb{N}$) = $\{1, 2, 3, 4, \dots\}$
2. **Positive integers** ($\mathbb{Z}^+$) = $\{1, 2, 3, 4, \dots\}$
3. **Negative integers** = $\{-1, -2, -3, -4, \dots\}$
4. **All integers** ($\mathbb{Z}$) = $\{\dots, -2, -1, 0, 1, 2, \dots\}$
5. **Even integers** = $\{\dots, -4, -2, 0, 2, 4, 6, \dots\}$
6. **Odd integers** = $\{\dots, -3, -1, 1, 3, 5, \dots\}$
7. **Multiples of any integer k** = $\{\dots, -2k, -k, 0, k, 2k, \dots\}$
8. **Integers divisible by 5 but not by 7**
9. **Integers not divisible by 3**
10. **Rational numbers** ($\mathbb{Q}$) = $\{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$
11. **Positive rational numbers** ($\mathbb{Q}^+$)
12. **Negative rational numbers** ($\mathbb{Q}^-$)
13. **Odd negative integers**
14. **Integers greater than 10**
15. **Prime numbers** = $\{2, 3, 5, 7, 11, 13, \dots\}$
16. **Perfect squares** = $\{1, 4, 9, 16, 25, \dots\}$
17. **Powers of 2** = $\{1, 2, 4, 8, 16, 32, \dots\}$
18. **Fibonacci numbers** = $\{1, 1, 2, 3, 5, 8, 13, \dots\}$

19. **Algebraic numbers** (roots of polynomials with integer coefficients)
 20. **Set $A \times \mathbb{Z}^+$ where $A = \{2, 3\}$** (Cartesian product with finite set)
 21. **All bit strings not containing the bit 0** $= \{1, 11, 111, 1111, \dots\}$
 22. **Finite strings over any finite alphabet**
 23. **Computer programs in any programming language**
 24. **All polynomials with integer coefficients**
 25. **Union of two countable sets**
 26. **Any subset of a countable set**
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UNCOUNTABLE SETS (Cannot be listed in a sequence)

Real Number Sets

1. **Real numbers ($\mathbb{R}$)**
2. **Real numbers between 0 and 1: $(0, 1)$**
3. **Real numbers between 0 and 2: $(0, 2)$**
4. **Real numbers between 0 and $1/2$: $(0, 1/2)$**
5. **Closed interval $[0, 1]$**
6. **Half-open interval $(0, 1]$**
7. **Irrational numbers ($\mathbb{R} - \mathbb{Q}$)**
8. **Transcendental numbers (e.g., π , e)**
9. **Real numbers not containing 0 in their decimal representation**
10. **Real numbers with only 1s or 9s in decimal representation**
11. **Positive real numbers ($\mathbb{R}^+$)**
12. **Negative real numbers ($\mathbb{R}^-$)**

Infinite Sequences and Functions

13. **Infinite sequences of 0s and 1s**
14. **Set of all functions from $\mathbb{Z}^+$ to $\mathbb{Z}^+$**
15. **Set of all functions from $\mathbb{R}$ to $\mathbb{R}$**

16. **Set of all infinite binary sequences**

17. **Uncomputable functions**

Power Sets

18. **Power set of natural numbers: $\mathcal{P}(\mathbb{N})$**

19. **Power set of integers: $\mathcal{P}(\mathbb{Z})$**

20. **Power set of any infinite set**

Geometric/Topological Sets

21. **Points in a line segment**

22. **Points in a square**

23. **Points in $\mathbb{R}^2$ (the plane)**

24. **Points in $\mathbb{R}^n$ (n-dimensional space, $n \geq 1$)**

25. **Points on a circle**

Set Operations with Uncountable Sets

26. **$\mathbb{R} - \mathbb{Z}$ (reals minus integers)**

27. **$\mathbb{R} - \mathbb{Q}$ (irrational numbers)**

28. **Union of uncountable set with any set (remains uncountable)**

29. **Any set containing an uncountable subset**

EXAMPLE PROOF 1: The Set of All Integers is Countable

Theorem

The set $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$ is countable.

Proof

Method 1: Explicit Listing

We can list all integers in a sequence by starting with 0 and alternating between positive and negative integers:

Sequence: 0, 1, -1, 2, -2, 3, -3, 4, -4, 5, -5, ...

This demonstrates that every integer appears exactly once in the sequence, so the set is countable.

Method 2: One-to-One Correspondence

We construct a bijection $f: \mathbb{Z}^+ \rightarrow \mathbb{Z}$ as follows:

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n-1)/2 & \text{if } n \text{ is odd} \end{cases}$$

Visual representation:

n (from $\mathbb{Z}^+$)	1	2	3	4	5	6	7	8	9	10	...
f(n) (to $\mathbb{Z}$)	0	1	-1	2	-2	3	-3	4	-4	5	...

Proof that f is one-to-one:

Suppose $f(m) = f(n)$. We consider cases:

Case 1: Both m and n are even.

- Then $f(m) = m/2$ and $f(n) = n/2$
- From $m/2 = n/2$, we get $m = n$ ✓

Case 2: Both m and n are odd.

- Then $f(m) = -(m-1)/2$ and $f(n) = -(n-1)/2$
- From $-(m-1)/2 = -(n-1)/2$, we get $m-1 = n-1$, so $m = n$ ✓

Case 3: One is even and one is odd.

- If m is even and n is odd: $f(m) = m/2 \geq 0$ but $f(n) = -(n-1)/2 \leq 0$
- These can only be equal if both are 0
- $f(m) = 0$ means $m = 0$, but 0 is not in $\mathbb{Z}^+$, contradiction
- Therefore this case is impossible ✓

Proof that f is onto:

Let $k \in \mathbb{Z}$ be arbitrary. We must find $n \in \mathbb{Z}^+$ such that $f(n) = k$.

Case 1: $k = 0$

- Choose $n = 1$ (odd)
- Then $f(1) = -(1-1)/2 = 0 = k \checkmark$

Case 2: $k > 0$

- Choose $n = 2k$ (even)
- Then $f(2k) = 2k/2 = k \checkmark$

Case 3: $k < 0$

- Choose $n = -2k + 1$ (this is odd since $-2k$ is even)
- Then $f(-2k + 1) = -((-2k + 1) - 1)/2 = -(-2k)/2 = k \checkmark$

Conclusion: Since f is both one-to-one and onto, it is a bijection from $\mathbb{Z}^+$ to $\mathbb{Z}$. Therefore, $|\mathbb{Z}| = |\mathbb{Z}^+| = \aleph_0$, and $\mathbb{Z}$ is countably infinite. ■

EXAMPLE PROOF 2: The Interval $(0, 1)$ is Uncountable

Theorem

The set of real numbers between 0 and 1, denoted $(0, 1) = \{x \in \mathbb{R} \mid 0 < x < 1\}$, is uncountable.

Proof (Cantor's Diagonalization Argument)

We will prove this by **contradiction**.

Assumption: Suppose $(0, 1)$ is countable.

Then we can list all real numbers in $(0, 1)$ as a sequence: $r_1, r_2, r_3, r_4, \dots$

Each real number has a decimal representation. Let:

$$\begin{aligned} r_1 &= 0.d_{11} d_{12} d_{13} d_{14} d_{15} \dots \\ r_2 &= 0.d_{21} d_{22} d_{23} d_{24} d_{25} \dots \\ r_3 &= 0.d_{31} d_{32} d_{33} d_{34} d_{35} \dots \\ r_4 &= 0.d_{41} d_{42} d_{43} d_{44} d_{45} \dots \\ r_5 &= 0.d_{51} d_{52} d_{53} d_{54} d_{55} \dots \\ &\vdots \end{aligned}$$

where each $d_{ij} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Concrete Example:

$$\begin{aligned} r_1 &= 0.2\,3\,7\,9\,4\,1\,0\,2\,\dots \\ r_2 &= 0.4\,4\,5\,9\,0\,1\,3\,8\,\dots \\ r_3 &= 0.0\,9\,1\,1\,8\,7\,6\,4\,\dots \\ r_4 &= 0.8\,0\,5\,5\,3\,9\,0\,0\,\dots \\ r_5 &= 0.1\,2\,3\,4\,5\,6\,7\,8\,\dots \\ &\vdots \end{aligned}$$

Construction: We now construct a real number $r = 0.d_1d_2d_3d_4d_5\dots$ using the following rule:

$$d_i = \begin{cases} 4 & \text{if } d_{ii} \neq 4 \\ 5 & \text{if } d_{ii} = 4 \end{cases}$$

What this means: For the i th decimal place of r , we look at the i th decimal place of r_i (the diagonal element d_{ii}) and choose a different digit.

Applying to our example:

- d_1 : Look at $d_{11} = 2 (\neq 4)$, so $d_1 = 4$
- d_2 : Look at $d_{22} = 4 (= 4)$, so $d_2 = 5$
- d_3 : Look at $d_{33} = 1 (\neq 4)$, so $d_3 = 4$
- d_4 : Look at $d_{44} = 5 (\neq 4)$, so $d_4 = 4$
- d_5 : Look at $d_{55} = 5 (\neq 4)$, so $d_5 = 4$
- ...

Result: $r = 0.45444\dots$

Key Observation: r differs from every number in our list:

- $r \neq r_1$ because they differ in the 1st decimal place ($d_1 \neq d_{11}$)
- $r \neq r_2$ because they differ in the 2nd decimal place ($d_2 \neq d_{22}$)
- $r \neq r_3$ because they differ in the 3rd decimal place ($d_3 \neq d_{33}$)
- $r \neq r_4$ because they differ in the 4th decimal place ($d_4 \neq d_{44}$)
- In general, $r \neq r_n$ because they differ in the n th decimal place ($d_n \neq d_{nn}$)

Important: Since we only use digits 4 and 5, r 's decimal expansion never ends in all 9s (which would create ambiguity like $0.5 = 0.4999\dots$).

Contradiction: We have constructed a real number $r \in (0, 1)$ that is NOT in our supposedly complete list. This contradicts our assumption that we could list all real numbers in $(0, 1)$.

Conclusion: Our assumption must be false. Therefore, $(0, 1)$ cannot be countable, so it is uncountable. ■

Why this works: No matter how we try to list the real numbers, we can always construct a new real number not in the list by systematically differing from each listed number in at least one decimal place. This shows there are "too many" real numbers to ever list completely.

CARDINALITY HIERARCHY

Finite sets (size n)



Countably Infinite ($\aleph_0$)



Uncountable (c for reals)



Larger infinities (2^c , etc.)

Key relationship: $\aleph_0 < c$

In words: There are strictly more real numbers than integers, even though both are infinite.

QUICK DECISION GUIDE

A set is COUNTABLE if:

- Can write as a list/sequence $a_1, a_2, a_3, \dots$
- Contains only integers (usually)
- Contains only rational numbers
- Is finite
- Is a subset of a countable set
- Can construct explicit bijection with $\mathbb{Z}^+$

A set is UNCOUNTABLE if:

- Contains an interval of real numbers (no matter how small)

- Contains "all possible" infinite sequences
 - Is a power set of an infinite set
 - Contains all functions from an infinite set to itself
 - Contains an uncountable subset
 - Cannot be listed (no matter how clever the listing strategy)
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KEY THEOREMS

1. **Union of countable sets:** If A and B are countable, then $A \cup B$ is countable
 2. **Subset of countable:** Any subset of a countable set is countable
 3. **Superset of uncountable:** Any set containing an uncountable subset is uncountable
 4. **Cartesian product:** $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable (surprising!)
 5. **Power set theorem:** For any set S, $|S| < |\mathcal{P}(S)|$ (Cantor's theorem)
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SURPRISING FACTS

1. $\mathbb{Z}$ and $\mathbb{Z}^+$ have the same size (both $\aleph_0$)
2. $\mathbb{Q}$ is countable despite being dense in $\mathbb{R}$
3. There are more irrational numbers than rational numbers
4. $(0, 1)$ and $\mathbb{R}$ have the same cardinality
5. Most functions from $\mathbb{Z}^+$ to $\mathbb{Z}^+$ are uncomputable (uncountably many functions, countably many programs)
6. $\mathbb{R}^2$ (the plane) has the same cardinality as $\mathbb{R}$ (the line)